



Aprendizagem 2025

Lab 5: Linear Regression

Practical exercises

1. Consider the following training data:

	y1	y2	output
x_1	1	1	1.4
x_2	2	1	0.5
x_3	1	3	2
x_4	3	3	2.5

- Find the closed form solution for a linear regression minimizing the sum of squared errors
- Predict the target value for $x_{new} = [23]^T$
- Sketch the predicted three-dimensional hyperplane
- Compute the MSE and MAE produced by the linear regression
- Are there biases on the residuals against y1? And y2?
- Compute the closed form solution considering Ridge regularization term with $\lambda = 0.2$.
- Compare the hyperplanes obtained using ordinary least squares and Ridge regression.
- Why is Lasso regression suggested for data spaces of higher dimensionality?

2. Consider the following training data where *output* is an ordinal variable.

	y1	y2	output
x_1	1	1	1
x_2	2	1	1
x_3	1	3	0
x_4	3	3	0

- Find a linear regression using the closed form solution
- Assuming the output threshold $\theta=0.5$, use the regression to classify $x_{new} = [22.5]^T$

3. Considering the following data

	y1	y2	output
x_1	3	-1	2
x_2	4	2	1
x_3	2	2	1

to learn a model:

$$z = w_1x_1 + w_2x_2 + \varepsilon, \text{ where } \varepsilon \sim N(0,5)$$

Compare:

- a) $w = [w_1w_2]^T$ using the maximum likelihood approach
- b) w using the Bayesian approach, assuming

$$p(w) = N\left(w \vee u = [00], \sigma = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix}\right)$$

4. Identify a transformation to aid the linearly modeling of the following data points.

	y1	y2	output
x_1	-0.95	0.62	0
x_2	0.63	0.31	0
x_3	-0.12	-0.21	1
x_4	-0.24	-0.5	0
x_5	0.07	-0.42	1
x_6	0.03	0.91	0
x_7	0.05	0.09	1
x_8	-0.83	0.22	0

Sketch the predicted surface.

5. Consider logarithmic and quadratic transformations:

$$\varphi_1(x) = \log(x) \text{ and } \varphi_2(x) = x^2$$

- a) Plot both of the closed form regressions.

	<i>input</i>	<i>output</i>
x_1	3	1.5
x_2	4	9.3
x_3	6	23.4
x_4	10	45.8
x_5	12	60.1

- b) Which one minimizes the sum of squared errors on the original training data?

6. Select the criteria that promotes a smoother regression model:

- a) Applying Lasso and Ridge regularization to linear regression models
- b) Increasing the depth of a decision tree regressor
- c) Increasing the k of a k NN regressor
- d) Parameterizing a k NN regressor with uniform weights instead of distance-based weights

Programming quest

7. Consider the *housing.arff* dataset and the *Regression* notebook available at the course's webpage.

Compare the determination coefficients of the non-regularized, Lasso and Ridge linear regression.